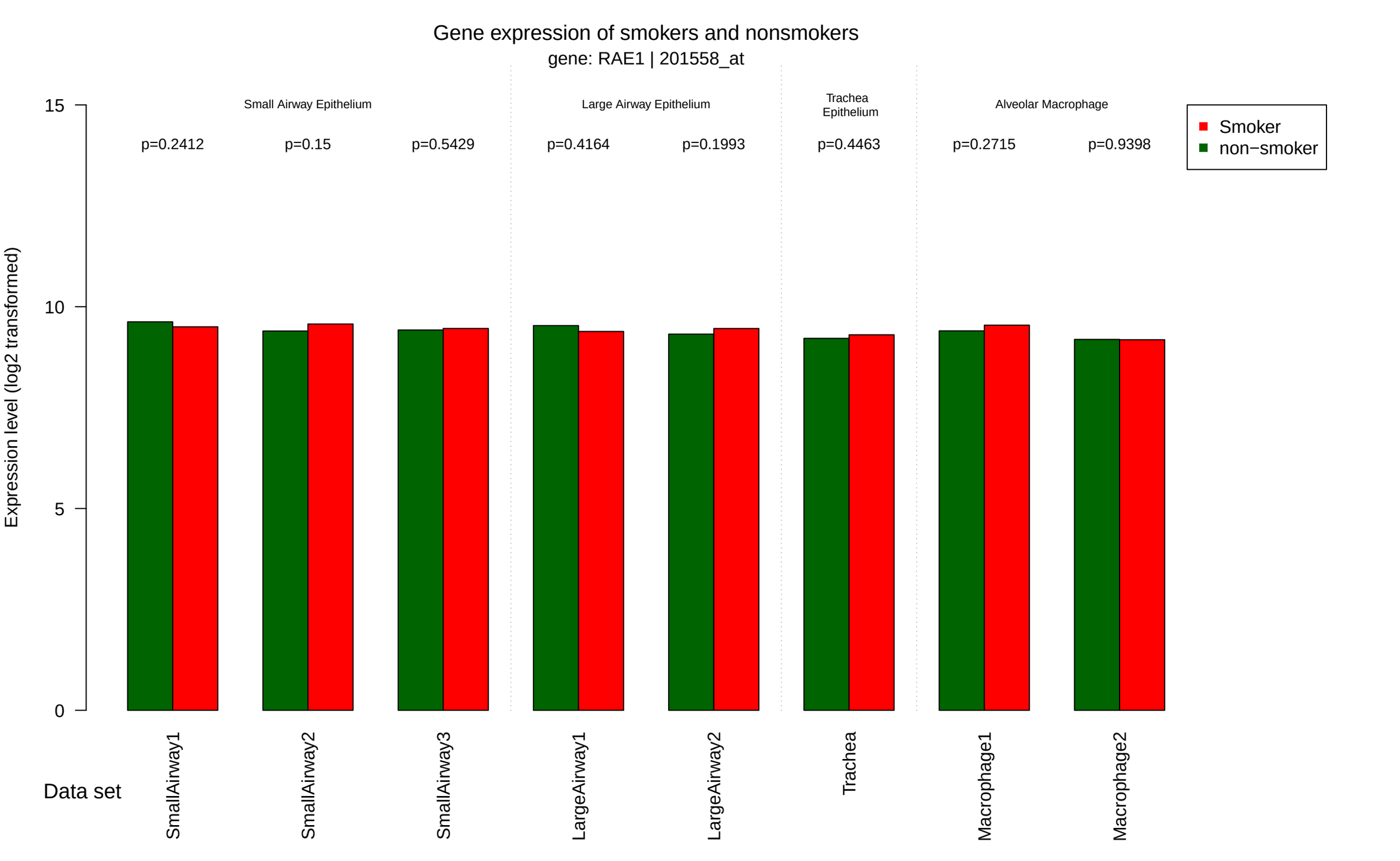
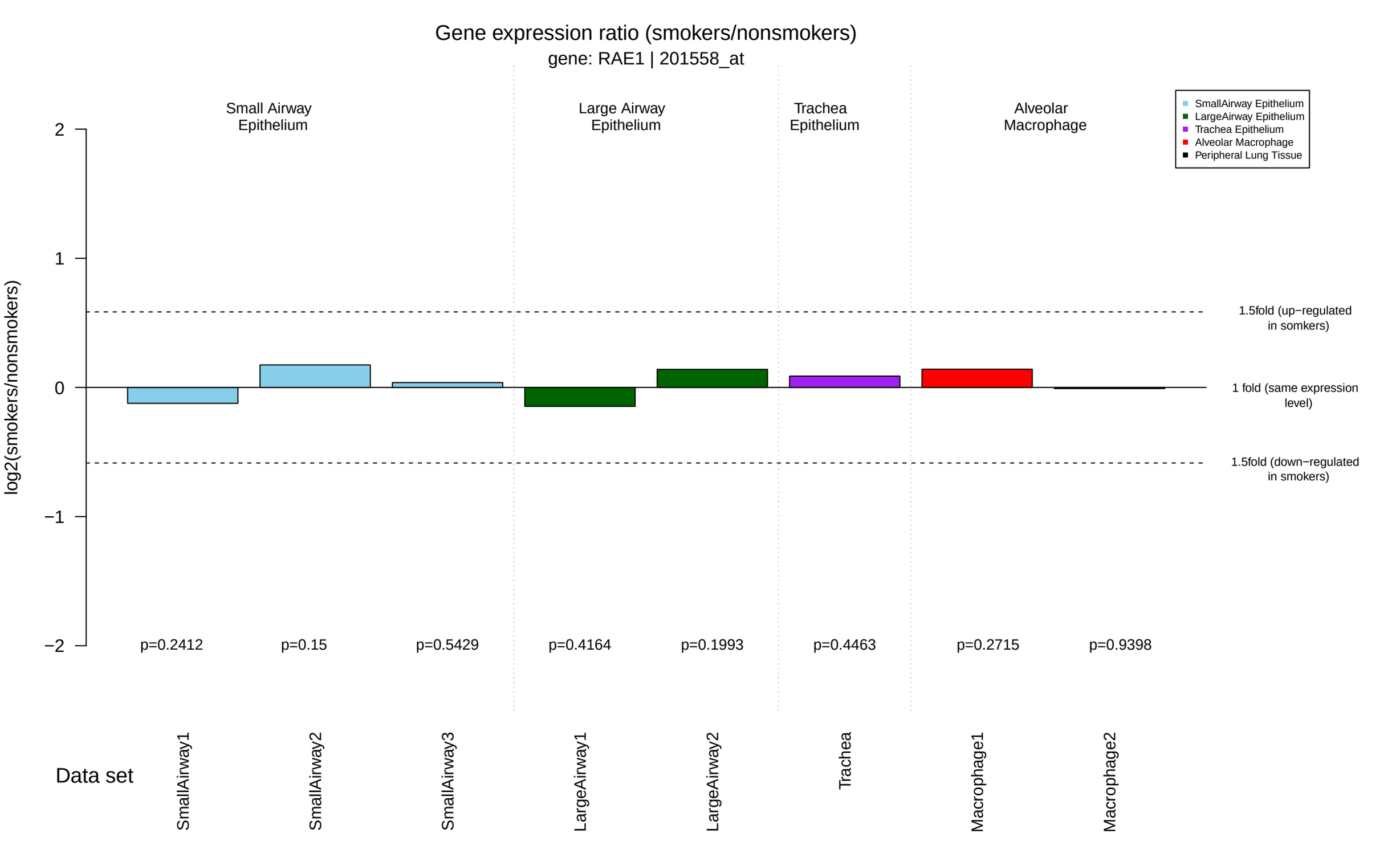
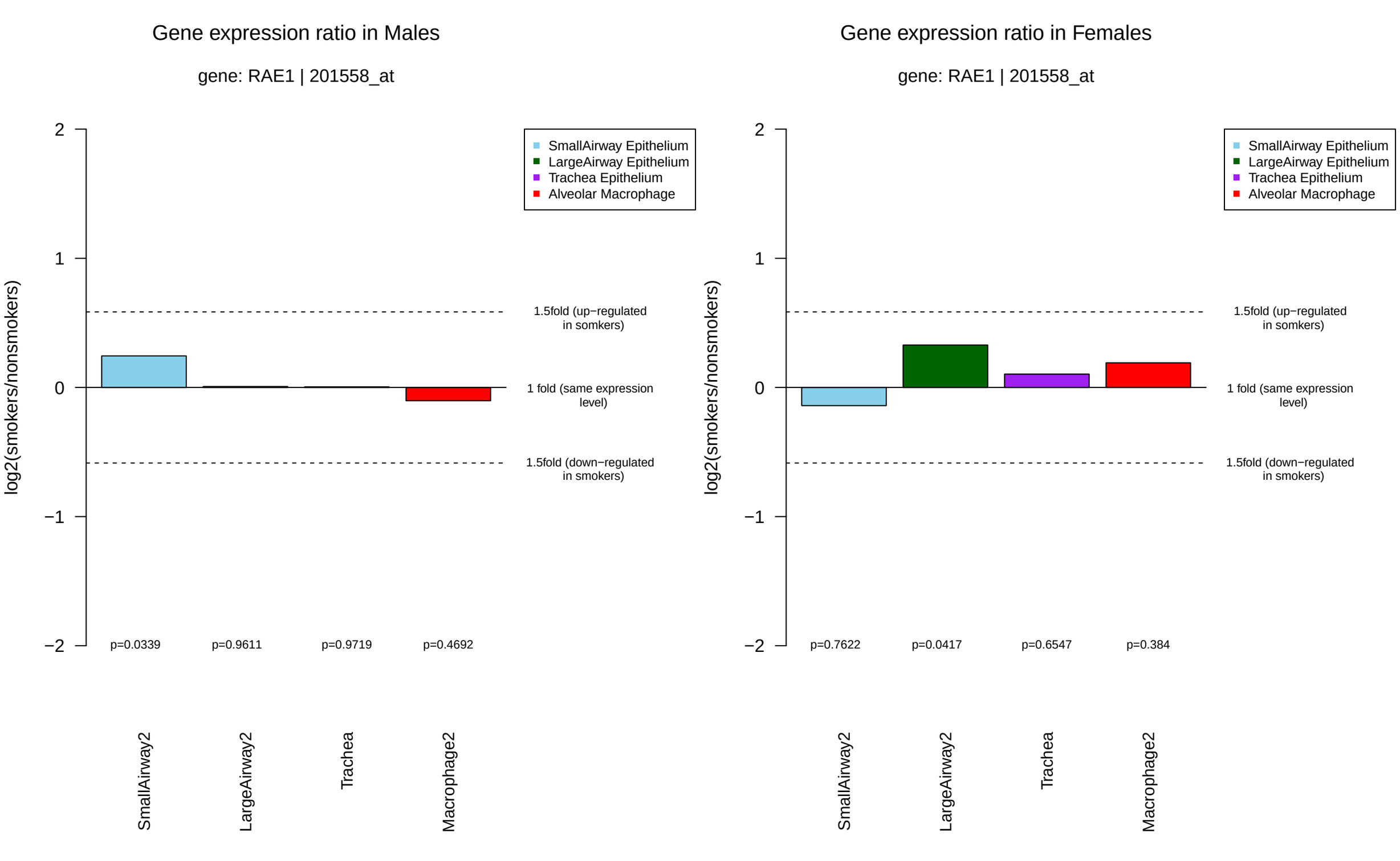
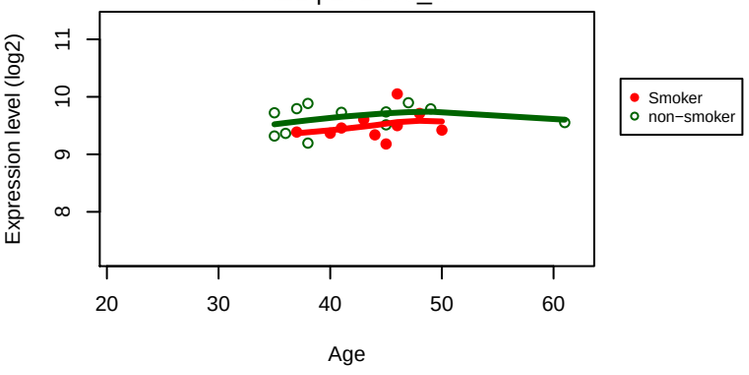
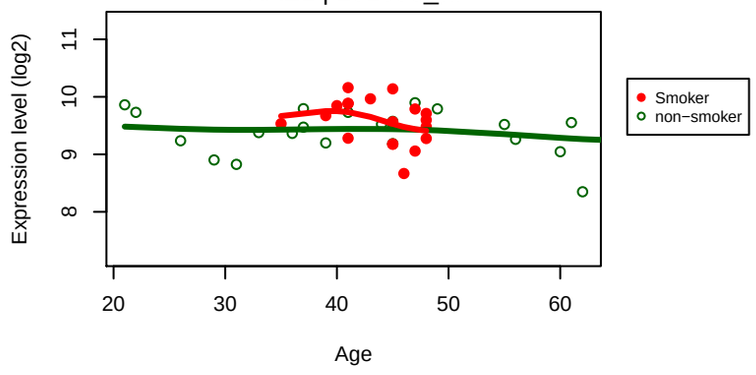
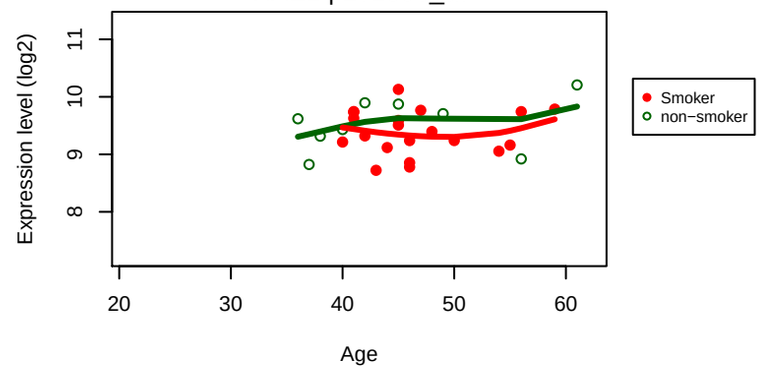

Gene expression of smokers and nonsmokers
gene: RAE1 | 201558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2412
p=0.15
p=0.5429
p=0.4164
p=0.1993
p=0.4463
p=0.2715
p=0.9398
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

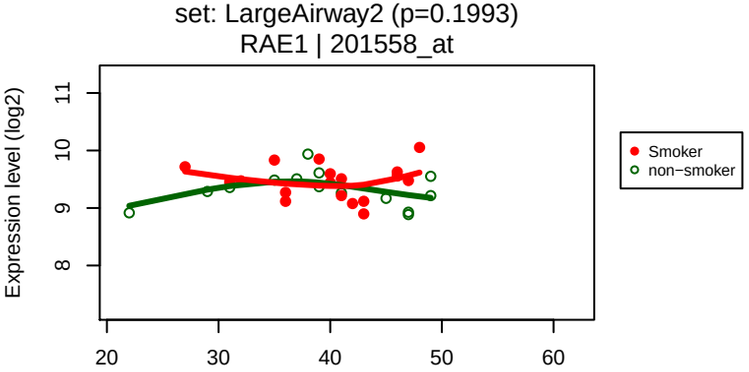

Gene expression ratio (smokers/nonsmokers)
gene: RAE1 | 201558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2412
p=0.15
p=0.5429
p=0.4164
p=0.1993
p=0.4463
p=0.2715
p=0.9398
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

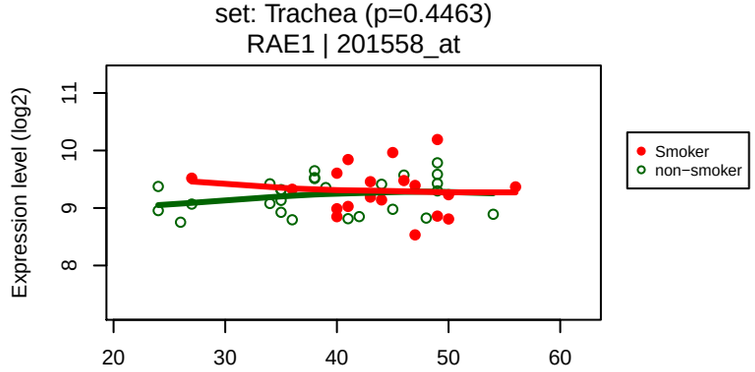

Gene expression ratio in Males
gene: RAE1 | 201558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0339
p=0.9611
p=0.9719
p=0.4692
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAE1 | 201558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7622
p=0.0417
p=0.6547
p=0.384
SmallAirway2
LargeAirway2
Trachea
Macrophage2

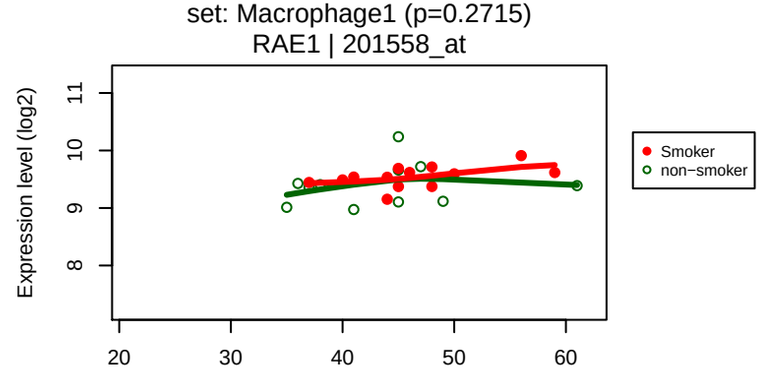

set: SmallAirway1 (p=0.2412)
RAE1 | 201558_at
Expression level (log2)
Age
Smoker
non−smoker

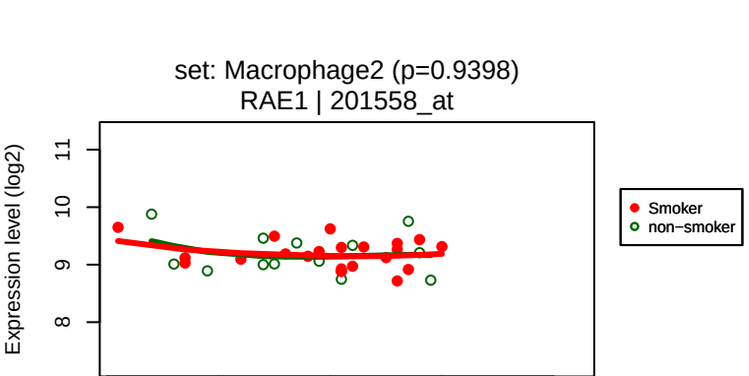

set: SmallAirway2 (p=0.15)
RAE1 | 201558_at
Expression level (log2)
Age
Smoker
non−smoker

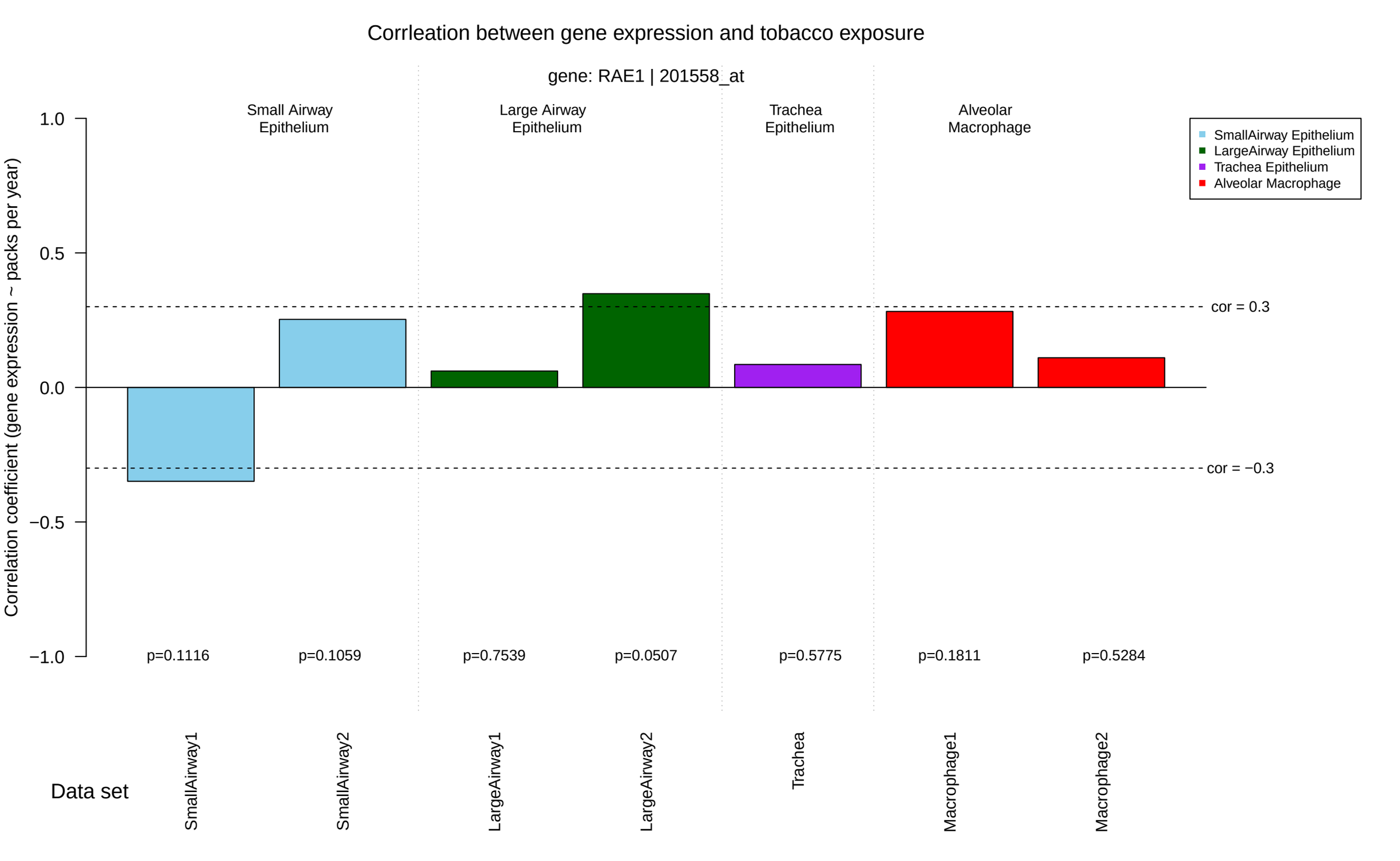

set: LargeAirway1 (p=0.4164)
RAE1 | 201558_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1993)
RAE1 | 201558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4463)
RAE1 | 201558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2715)
RAE1 | 201558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9398)
RAE1 | 201558_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAE1 | 201558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1116
p=0.1059
p=0.7539
p=0.0507
p=0.5775
p=0.1811
p=0.5284
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2